Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



# Team Details

a. Team name: **Pivot Analytix**
b. Team leader name: **Abdullah Moazzam**
c. Team members: **Saad Wali Muhammad Joiya, Usama Bin Maalik**
d. Problem Statement: **Theme 5 · Future of Work & Enterprise Productivity.** Designing a home takes weeks of manual drafting. Checking it against building bylaws is done by hand, by experts in short supply. And most families commit to building from 2D drawings they can't picture. Naqsha's AI agents turn a simple conversation into a bylaw-compliant 3D home you can walk through, in minutes. No product does this today. Naqsha is set to transform how architects and residential construction companies work, and how families decide what to build.

# Brief about the idea

Theme 5 · Future of Work & Enterprise Productivity

## From a conversation to a bylaw-compliant 3D home you can walk through, in minutes.

No product offers this today: we are redefining how families and architects design a home. Naqsha (نقشہ) is Urdu for “house plan”.

### The problem

- **Design takes weeks.** Every draft and every change is drawn by hand.
- **Most families never see their home in 3D,** let alone walk through it. They commit to building from flat drawings.
- **Compliance needs scarce experts.** They check every bylaw by hand, and mistakes often show up only when the authority refuses the plan.
- **Feedback is scattered.** Owners, family, architects and builders give it in different places, so every round of changes is slow.




### The solution

**Naqsha’s AI agents: an experienced architect you can talk to.**

- **Designs in minutes, by conversation.** Describe the plot and the family in plain words. Naqsha drafts the concepts, the floor plans and the full 3D home.
- **Compliant from the start.** Every plan is checked against the local authority’s bylaws, and Naqsha fixes what fails.
- **Immersive, high-fidelity 3D.** Walk through the furnished, finished home, or view it from any angle, by day or night.
- **Comments that get addressed.** Pin a note anywhere as you walk. Naqsha makes the change, or names the rule that stops it.
- **One project for everyone.** Owners, family, architects and builders review and comment together.
- **Change it by asking.** “Make the kitchen bigger” gives you a new, checked version.

Prototype: Lahore (LDA) bylaws. Other authorities’ rules load the same way.

# Opportunities

Compared with how homes are designed today

## How different is it from any of the other existing ideas?

**No product today takes families and their architects from a conversation to a compliant 3D home they can walk through, comment on and change.**

- **Architect or draftsman:** weeks per design, in 2D, with every change redrawn by hand.
- **3D renders:** ordered separately, they take days and are still pictures you can't walk through.
- **AI image and floor-plan tools:** a picture or a 2D layout, with no bylaw checks, no walkthrough and no shared project.

## How will it be able to solve the problem?

- **Weeks of drafting:** a complete design in minutes, from one conversation.
- **Never seen in 3D:** every design is a high-fidelity 3D home you can walk through.
- **Scarce compliance experts:** bylaw checks are built in, so the architect stamps a set that is already checked.
- **Scattered feedback:** comments pinned in the model and addressed by the AI architect, one shared project, and changes made by asking.

## USP of the proposed solution

**An experienced architect you can talk to.**

- AI agents that design the whole home in minutes, check their own work against the bylaws and fix it.
- Immersive, high-fidelity 3D, inside and out, in the browser.
- Comments pinned where you stand, and addressed for you.
- Collaboration and changes in plain language, all in one place.
- Rules as data: LDA in the prototype, and any authority's bylaws can be added.

# List of features offered by the solution

**From your requirements to a complete, high-fidelity, bylaw-compliant 3D walkthrough of your home, in minutes.**

### Design with an AI architect
Describe your plot, family and wishes in plain words; Naqsha asks only for what's missing.

### Concepts by day and night
Four photoreal exterior options for your plot, each also shown at night.

### Compliant floor plans
Every floor is checked against the authority's bylaws and livable-home rules; Naqsha fixes what fails and reviews its own drawing.

### High-fidelity 3D, inside and out
Furnished rooms, finished facades, any angle, and cut-away floors.

### Immersive walkthrough
Walk room to room, jump to any room, and take the stairs up to the roof.

### Night lighting and photoreal renders
Light the house at night, or turn any view into a photoreal image in seconds.

### Comments where you stand
Pin a note on any wall, window or room as you walk; everyone on the project sees it.

### Comments addressed for you
The AI architect makes each change, or names the rule that stops it and offers another way.

### Change it by asking
Type a request in plain language and get a new, checked version.

### Collaborate on one project
Owners, family, architects and builders together; client links need no account.

### Versions and drawings
Every version is kept with its changes in feet, plus a 1:100 print set and a compliance report.

### Bylaws for your area
LDA in the prototype; other authorities' rules load as data.

**Who it helps**

**Families** see, walk and shape their home before they build.

**Architects** deliver a checked design in the first meeting.

**Construction companies** sell and build from designs buyers can walk through.

# Process flow diagram or Use-case diagram

| 1 Brief | 2 Concepts | 3 Plan | 4 3D house | 5 Walk & share | 6 Revise |
|---|---|---|---|---|---|
| ~3 s a turn | ~20 s for 4 | 4.6–5.2 min for a 10 marla home | | instant | ~70 s a version |
| Describe the plot in one message. Google ADK agent · Gemini 3.8 Flash | Choose a photoreal look, day or night. Gemini 3.1 Flash Image | A plan the agent checks and fixes. Gemini 4 Argon loop A ↓ | Watch the house build, part by part. Writes it as code loop A ↓ | Walk it like a game; share a link. three.js; Gemini renders any view | Pin comments, get the next version. Google ADK agent loop B ↓ |

## Loop A — Architect pipeline · Gemini 4 Argon

steps 3–4 · checks itself before you see it

critique → revise → re-check

**Plan** every floor as JSON → **Rules check** a tool call: 36–43 LDA + 10 livable → **Self-review** looks at its own drawings → **Write code** design.js, ~25 parts, streamed → **As-built** setbacks measured on the 3D model

fails → fix → re-check (15 → 5 → 0)

**Rules engine:** each authority's bylaws as deterministic code (LDA 2026 in the prototype), plus **10 livable-home rules**. The same engine draws the envelope (1) and dry-runs edits (6).

tool use · self-correction · multimodal self-review · code generation · rules as guardrails

## Loop B — Revision agent · Gemini 4 Argon

Agentic harness: Google ADK · step 6 · 9 tools

1. Reads the comments and their 3D snapshots
2. Opens only the rooms and code it needs
3. Dry-runs each edit against the rules

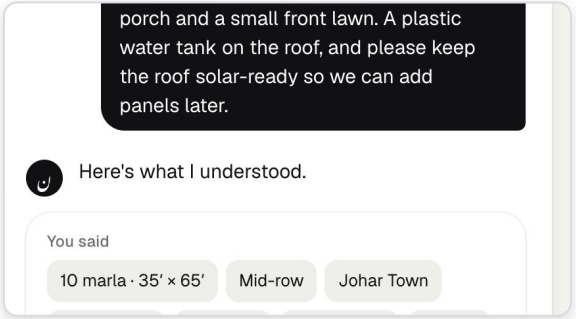
**Passes:** rewrites those parts and answers the comment

**Breaks a rule:** names it and offers an alternative

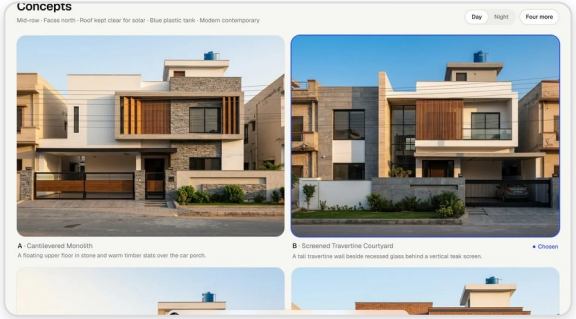
4. Owner picks what goes in: next version

tool use · refuses with the rule · human approval

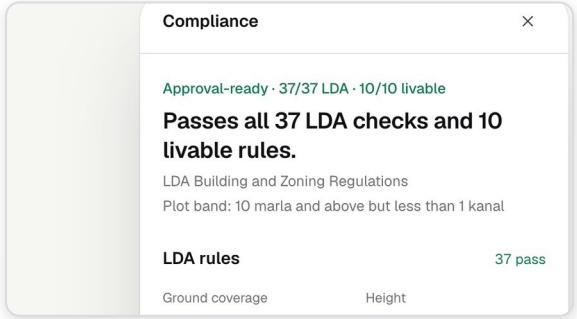
# Wireframes/Mock diagrams of the proposed solution (optional)




**1 Brief:** agent asks what's missing



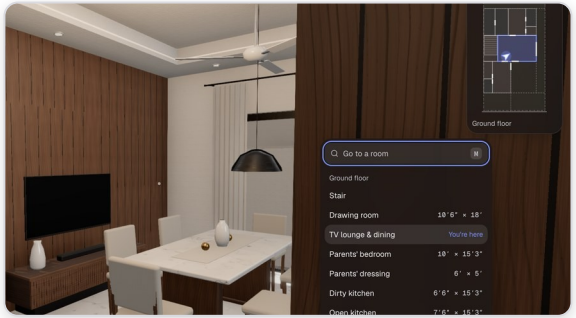

**2 Concepts:** 4 looks, day or night



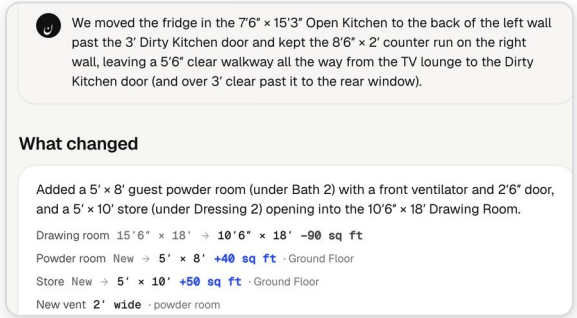

**3 Plan:** agent-checked, 37/37 LDA



**4 House:** coded by Gemini 4 Argon




**5 Walk:** first person, room finder

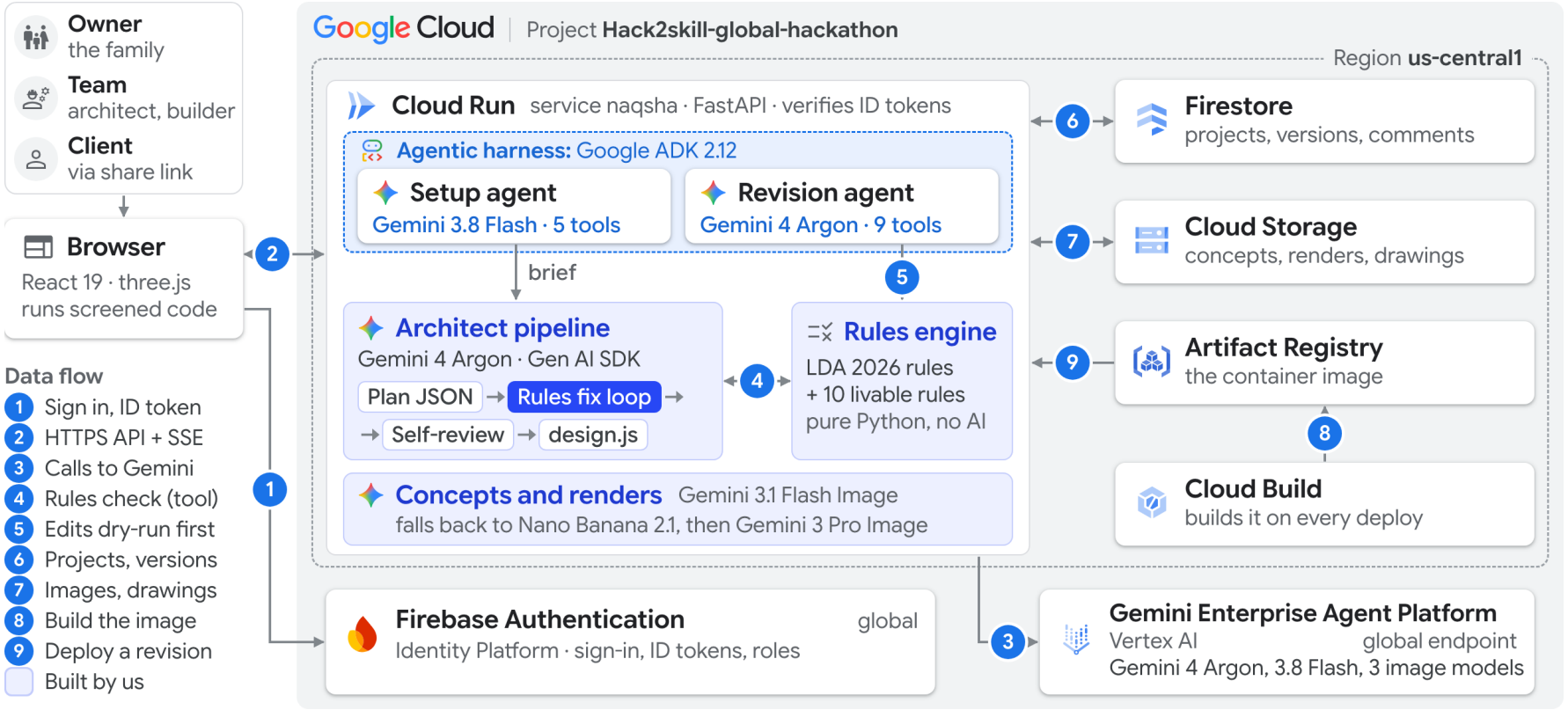



**6 Revise:** agent lists changes in feet

- All six are live screens captured from the deployed app on Cloud Run; the crops show the panel each step is about.

# Architecture diagram of the proposed solution

● Live on Cloud Run · us-central1




**Data flow**

1. Sign in, ID token
2. HTTPS API + SSE
3. Calls to Gemini
4. Rules check (tool)
5. Edits dry-run first
6. Projects, versions
7. Images, drawings
8. Build the image
9. Deploy a revision

Built by us

# Technologies to be used in the solution

Google platform | Open source | Built by us

## Agentic harness: Google ADK 2.12 · 2 agents, 14 tools

**Setup agent** Gemini 3.8 Flash · 5 tools **Revision agent** Gemini 4 Argon · 9 tools both run in the Cloud Run service

### Agent Platform — Vertex AI

- **Gemini 4 Argon** plan, 3D code, revise
- **Gemini 3.8 Flash** setup agent
- **Gemini 3.1 Flash Image** concepts
- **Nano Banana 2.1** image fallback 1
- **Gemini 3 Pro Image** image fallback 2
- **Global endpoint** newest models only

### Google Cloud — one project

- **Cloud Run** API, React app, agents
- **Firestore** projects, versions
- **Cloud Storage** images, drawings
- **Firebase Authentication** sign-in, roles
- **Cloud Build** builds each deploy
- **Artifact Registry** container images

### Architect pipeline — Gemini 4 Argon

- **google-genai 2.31** Gen AI SDK
- **Rules check (tool)** LDA + livable rules
- **Rules fix loop** until every check passes
- **Multimodal review** its own drawings
- **Resilience** 6 tries, 4–60 s backoff

**Fixed-step agent loop:** ADK not needed.

### Backend — api/

- **Python 3.13** uv for packages
- **FastAPI** Server-Sent Events
- **Pydantic** API models
- **Pillow** plan drawings
- **pytest** 123 tests
- **Docker** one image: web + API

### Frontend and 3D — web/

- **React 19** TypeScript, Vite
- **Tailwind CSS 4** app styling
- **three.js r186** WebGL renderer
- **react-three-fiber + drei**
- **three-mesh-bvh** walk physics
- **N8AO** ambient occlusion

### Built by us — domain IP

- **House SDK** 44 materials
- **LDA rules as data** 2026 regulations
- **10 livable rules** beyond the bylaws
- **As-built check** setbacks in 3D
- **1:100 print sheets** A3, one per floor
- **Code screen** 21 banned patterns

# Estimated implementation cost (optional)

## ≈ $2.65 of cloud and AI per new home

A 10 marla home, with Gemini 4 Argon at its announced 1 Dec 2026 prices (conservative), about Rs 740. At today's prices: ≈ $1.60, about Rs 450.

| | | |
|---|---|---|
| **Setup chat, ~6 turns**<br>Gemini 3.8 Flash · ~60k in, 6k out |  | **$0.14** |
| **4 concepts + 4 night views**<br>Gemini 3.1 Flash Image · 8 images |  | **$0.54** |
| **Plan, fix loop, self-review**<br>Gemini 4 Argon · ~60k in, 50k out |  | **$1.24** |
| **3D house as code**<br>Gemini 4 Argon · ~25k in, 30k out |  | **$0.70** |
| **Hosting and storage**<br>Cloud Run ~6 min · Firestore · Storage |  | **$0.03** |
| **Each revision**<br>ADK + Gemini 4 Argon · ~120k in, 15k out · ≈ Rs 220 | | **≈ $0.78** |

Token counts estimated from prompt and output sizes. Rs 280 = US$1.

All figures are estimates from list prices

## ≈ 55 homes

cost the same as one avoided LDA resubmission: the Rs 41,430 scrutiny fee (Rs 15/sq ft × 2,762 sq ft, LDA 2026, Appendix C)

### Pilot month 1,000 homes + 3,000 revisions

| | |
|---|---|
| Gemini on Agent Platform | ~$5,000 |
| Cloud Run, one warm instance | ~$105 |
| Firestore + Cloud Storage | < $50 |
| **Total** | **≈ $5,200** |

### Pricing tiers to test prices set in pilots

- **Free** one home; samples, no account
- **Family** per home: revisions, client links, 1:100 set
- **Professional** architects and builders: team seats
- **Stamping partners** PCATP architects; referral fee

# Snapshots of the prototype

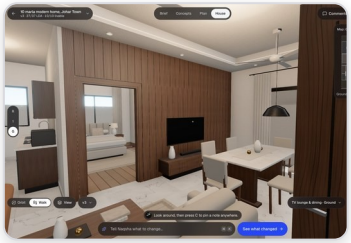
● All captured from the live service




**The 1 kanal Spanish villa** (43/43 LDA · 10/10 livable), written as 3D code by the AI architect (Gemini 4 Argon): tiled roofs, rooftop solar, a two-car porch. Neighbours turn see-through as the camera passes.




**Render this view:** Gemini turns any 3D view into a photo, same camera, in ~10 s.



**Walk mode: TV lounge**



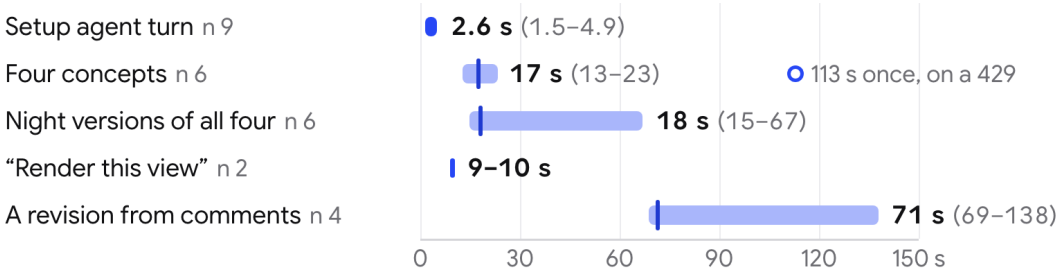
**1:100 print set**

# Prototype Performance report/Benchmarking

## Time to result, measured live

Cloud Run request logs · revisions 00007–00009 · 17 Oct 2026 · n = runs

**Interactive steps** · seconds · bar = range, tick = median




**Full design, plan to walkable 3D** · minutes · includes fix loop and self-review




**End to end, 10 marla:** 5.3 min of AI time from the first message to a walkable home (setup 3 s, concepts 23 s, night 17 s, plan to 3D 4.6 min). Today: weeks.

## Rules passed

Deterministic checks on what the agents delivered

**10 marla modern home**
**37/37** LDA
**10/10** livable · 0 warnings

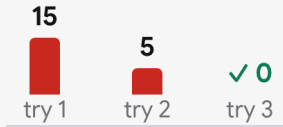
**1 kanal Spanish villa**
**43/43** LDA
**10/10** livable · 0 warnings

**8 → 0** Rule failures fell from a median of 8 on the first try to 0 delivered, fixed by the AI architect

All first designs saved in Firestore, 16–17 Oct.

**Loop A at work**

Rule failures per try on a 1 kanal brief; the AI architect fixed them.




**For scale:** an AI floor-plan tool we tested took ~6 min and 8 actions to draw one floor in 2D, with no stairs and no rule checks.

**~200** issues found and fixed in development by 8 AI tester agents over 3 rounds

**123** API tests, incl. rules engine

**6 tries** backoff on 429 and 5xx

**3** image models in the fallback chain

# Additional Details/Future Development (if any)

## Next weeks

- Drag rooms with an instant bylaw re-check
- Google sign-in
- Generated code in a sandboxed iframe
- glTF export, token cost dashboard

## Scale months

- Full rule packs for DHA, Bahria, RDA, FDA, MDA and GDA, then authorities beyond Punjab
- Pilots with PCATP architects and Lahore builders
- Sets ready for online submission, starting with LDA's

## Later ideas

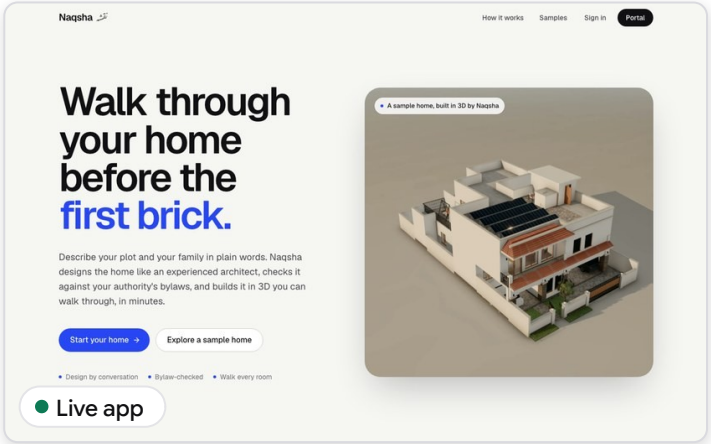
- An Urdu interface
- Cost and material estimates
- Structural and services drafts
- Rule packs for cities outside Pakistan

## Trust and responsibility

- **Pre-check only:** a PCATP-registered architect (Pakistan Council of Architects and Town Planners) stamps every plan.
- **Safe code:** Gemini's design.js is screened for 21 banned patterns (network, eval, DOM, storage) and runs only in the viewer's browser.
- **Deterministic rules:** each authority's official bylaws, written as code and tested; the prototype uses LDA's 2026 regulations.
- **People decide:** the owner picks which comments become changes.

## Provide links to your:

- GitHub Public Repository
  github.com/abdm693/hack2skill-hackathon
- Demo Video Link (3:57)
  https://drive.google.com/file/d/1xmdn7qxZRsblvwNLezveUji1tdc8WjBB/view?usp=sharing
- Final Product Link
  naqsha-242926648187.us-central1.run.app




**Try it in a minute:** open a sample home, no account.

**Repo guide** README (run and deploy) · docs/ARCHITECTURE.md · contracts/ (plan schema, House SDK, API) · api/tests (123 tests)

**Video** a 4-minute walkthrough of the live app.

**Scan to open the live app**

Cloud Run · us-central1

## Track and industry impact

Track · Theme 5 · Future of Work & Enterprise Productivity

# Future of Work in AEC: weeks of redrawing become one agentic session

Industry · Architecture, Engineering & Construction (AEC) + residential real estate

Pakistan first, then worldwide · Lahore prototype

**SPEED**
**~5–6 min**
of AI time, first message to a walkable, bylaw-checked home

**COST TODAY**
**Rs 41,430**
fee paid again when a 10 marla plan is refused in Lahore

**ITERATION**
**~70 s**
median, from comments to a checked version, with reasons

**MARKET**
**≈10 M**
homes short in Pakistan; first city, Lahore, has 13.0 M people

### Family — Owner

**Today:** a paper sketch, a contractor's guess and weeks of back-and-forth.

**With Naqsha**
- Describe the plot once
- Pick a look, by day or night
- Walk the house before it is built
- Choose which comments go in

### Architect, builder — Team

**Today:** redraws every change by hand; rule breaks surface after submission.

**With Naqsha**
- A checked, walkable design at the first meeting
- Agents revise from comments, ~70 s
- Print the 1:100 set + compliance

### Client, relatives — Share link

**Today:** waits for printouts and site visits to see the house.

**With Naqsha**
- Open a link, no account needed
- Walk the house on a phone
- Comment by name; the agent answers
- Approve a version

### PCATP architect — Partner · next

**Today:** stamps plans that may still break bylaws; a refusal means resubmission.

**With Naqsha**
- Receives a bylaw pre-checked 1:100 set
- Reviews, stamps and submits to the authority
- Fewer refusals and repeat fees

**Terms:** LDA = Lahore Development Authority · PCATP = Pakistan Council of Architects and Town Planners · 10 marla ≈ 2,250 sq ft (209 $m^2$) · 1 kanal ≈ 4,500 sq ft (418 $m^2$) · Rs 280 ≈ US$1
**Sources:** fee: LDA Building & Zoning Regulations (09-02-2026), App. C, Rs 15/sq ft × 2,762 sq ft covered · Lahore district: Census 2023 (PBS) · housing shortfall: World Bank estimate

Google Cloud H2S

# AI Builder Cup

Build in JAPAC.
Build for the world.

# Thank you

## Naqsha نقشہ

Walk through your home before the first brick.

**Pivot Analytix** · AI Builder Cup 2026

**Live app** naqsha-242926648187.us-central1.run.app

**Code** github.com/abdm693/hack2skill-hackathon

